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Improvement in HDL and lowering of LDL blood cholesterol level by feeding dietary chitosan extracted from exoskeleton of freshwater crab *Sartoriana spinigera* to albino rats.

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Abstract: Presently, hypercholesterolemia is one of the biggest health issues in human population leading to various cardiovascular diseases. In this experiment, a zootherapeutic medicine, chitosan, was extracted from a freshwater crab, *Sartoriana spinigera* locally found in Jharkhand and compared with cellulose and a hypolipidemic drug Ezetimibe to see hypocholesterolemic effect on LDL and HDL level of blood cholesterol in hypercholesterolemic albino rats. Experiment was conducted on 15 albino rats for 30 days. They were divided into 3 groups A,B and C and were given respective diets:-Group A(high fat diet+5% cellulose), group B(high fat diet+5% chitosan) and group C(high fat diet+5% hypolipidemic drug Ezetimibe).Blood of rats was collected on initial and final day to measure the level of HDL and LDL all groups. Statistical analysis by Student's t test between the difference of group A and group B showed that in group B HDL level was increased significantly($p<0.01$),whereas analysis between group B and C showed significantly increased concentration of HDL in group B($p<0.05$). Statistical analysis for LDL level in blood between group A and B showed that in group B, LDL was decreased significantly ($p<0.001$), whereas, statistical analysis between group B and C showed significantly decreased LDL level in group B ($p<0.001$), indicating that chitosan has the capacity to increase HDL (good cholesterol) and decrease LDL (bad cholesterol) in blood and is more potent hypocholesterolemic agent than cellulose and the reference drug in this study.

Keywords : Hypercholesterolemia, Zootherapeutic medicine, Cellulose, Ezetimibe, Students t test

INTRODUCTION

Hypercholesterolemia has been identified as a key risk factor for the development of various liver and cardiovascular diseases. Hypercholesterolemia results due to bad food habits, high fat intake in junk foods, etc. Increased cholesterol and LDL level in the body leads to visceral fat deposition which may seem to be an inert lump of lard, but it is actually highly active and constantly pumping poisons into the bloodstream. Increased level of LDL accumulates in the artery wall where they are oxidized and taken up by foam cells leading to development of atherosclerosis.

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Zootherapy has traditionally been employed in the treatment and management of different ailments and diseases. Chitosan is one such zootherapeutic medicine. It is a deacetylated form of chitin found in the exoskeleton of marine and freshwater crustaceans, molluscs and cell wall of fungi. It is a non toxic nutritional supplement generally regarded as a safe compound (Thanou *et al* 2001)[1].It is a natural polysaccharide ,comprising copolymers of glucosamine{â(1-4) linked 2-amino-2-deoxy-D-glucose} and N-acetyl-glucosamine(2-acetamido-2-deoxy-D-glucose) which can be achieved by partial deacetylation of chitin (Baldrick 2010)[2]. Chitosan is known to have many medicinal properties such as antibacterial, antiobesity, hypolipidemic properties (Gades

et al, 2003)[3]. But very few research work has been done on chitosan extracted from freshwater crabs. In this experiment, chitosan was extracted from freshwater edible crab *Sartoriana spinigera* which is a local crab of Jharkhand found abundantly in ponds, paddy fields during rainy season.

Earlier to this chitosan was extracted from shells of *S.spinigera* of Bangladesh by Muslim *et al* (2014)[4] to study the iron adsorption behaviour of some chitosan derivatives. *S.spinigera* is not only a local delicacy of Jharkhand but is also used as a medicine against various health problems. Hence the objective of this research was to extract chitosan from freshwater crab *Sartoriana spinigera* and to study its effect on HDL and LDL blood cholesterol level in hypercholesterolemia induced albino rats, and compare it with a plant source, cellulose and a synthetically produced hypolipidemic drug Ezetimibe.

MATERIALS AND METHODS

Fibres:

In this experiment, three types of fibres were used i.e.,Cellulose, Chitosan and a hypolipidemic drug Ezetimibe. Ezetimibe was purchased commercially while chitosan was extracted from exoskeleton of freshwater crab *Sartoriana spinigera* following the method of Takiguchi (1991)[5].

Extraction of chitosan:

Following the method of Takiguchi (1991)[5],the carapace powder of *Sartoriana spinigera* was undergone through three processes namely demineralization with 2N HCl for 24 hours, deproteinization with 1N NaOH at 80°C to obtain chitin and deacetylation of chitin in 50% NaOH at 100°C to obtain chitosan.

Animals and diet:

15 albino rats (*Rattus albus*) each measuring about 80 g were purchased from a rearing laboratory. They were randomly divided into 3 groups A, B and C and housed in metallic cages under healthy condition at room temperature. Rats were fed with respective diets at 9 am and 9 pm. The rats were acclimatized for 7 days with basal diet. Experimental diets were given for 30 days.

Group A – high fat diet + 5% cellulose (cabbage leaves)
Group B – high fat diet + 5% chitosan
Group C – high fat diet + 5% hypolipidemic dug Ezetimibe

Collection of blood samples:

The animals were fasted for 24 hours before the first experimental dose and after the last dose of extract, cellulose and reference drug. Blood was obtained by retro-orbital puncturing using a Corning glass micro hematocrit capillary tube of 1.2mm inner diameter after anesthetizing the rats with ether, and collecting the blood in clean bottles without anticoagulant ,to test the HDL and LDL blood cholesterol level on 1st(initial) and 30th (final) day.

Statistical analysis:

All values were expressed as the mean ±SD of 5 animals. The significance of difference between group A and B, between B and C, and between A and C was determined by Student's t test. Differences were considered significant at $p<0.05$.

Table 1:- Composition of experimental diet

Component	Group A High fat diet + 5% cellulose (g/1000g food)	Group B High fat diet + 5% chitosan (g/1000g food)	Group C High fat diet + 5% ezetimibe (g/1000g food)
Casein	147	147	147
Protein	306.6	306.6	306.6
Sucrose	140	140	140
Soyabean oil	15	15	15
Vitamin + mineral mix.	35	35	35
Cholesterol	5	5	5
Deoxycholic acid	2	2	2
Coconut oil	300	300	300
Cellulose	50	-	-
Chitosan	-	50	-
Hypolipidemic drug ezetimibe	-	-	50

RESULTS & DISCUSSION

Table 2:- Effect of dietary fibre Chitosan, cellulose and hypolipidemic drug on HDL and LDL blood cholesterol (mg/dL) levels of hypercholesterolemic albino rats.

Groups with different fibres	HDL cholesterol			LDL cholesterol		
	Initial (mg/dL)	Final (mg/dL)	Difference (%)	Initial (mg/dL)	Final (mg/dL)	Difference (%)
Group A (5% cellulose) Vs **Group B** (5% chitosan)	27±1.2 30±1.7	20±1.6 36±1.2	-25.92 20** *	36±1.5 37±1.3	39±1.6 30±2.4	8.33 -18.91***
Group B (5% chitosan) Vs **Group C** (5% ezetimibe)	30±1.7 26±2.2	36±1.2 29±1.7	20* 11.53	37±1.3 37±1.15	30±2.4 38±1.17	-18.91*** 2.70
Group A (5% cellulose) Vs **Group C** (5% ezetimibe)	27±1.2 26±2.2	20±1.6 29±1.7	-25.92 11.53***	36±1.5 37±1.15	39±1.6 38±1.17	8.33 2.70

*=significant at 5% level, **=significant at 1% level, ***=significant at 0.1% level

Table 2 showed the mean values of HDL and LDL levels on initial and final experimental day of all the three groups and the percentage change in their values. Group A fed with high fat diet and 5% cellulose showed a 25.92% decrease in HDL. Group B that was fed with high fat diet and 5% chitosan showed maximum level of HDL i.e. 20% increase was reported on the final day. Group C that was fed high fat diet and 5% hypolipidemic drug Ezetimibe showed only 11.53% increase in HDL.

Table 2 also showed that in group A LDL level was increased by 8.33% but in case of group B LDL was decreased by 18.91%, whereas in group C, only 2.70% LDL level in blood was increased.

Statistical analysis by Student's t test was performed between the difference of group A and group B. In group B treated with chitosan significantly increased level of HDL was observed($p<0.001$),whereas statistical analysis between group B and C showed significantly increased concentration of HDL in group B fed with chitosan than in group C that was given hypolipidemic drug at 5% level.

Comparison for LDL level in blood between group A and B showed that in group B,LDL was decreased significantly ($p<0.001$), whereas ,statistical analysis between group B and C showed significantly decreased LDL level in group B ($p<0.001$).

The above result indicates that group B fed with chitosan showed best HDL and LDL blood cholesterol conditions as compared to the groups treated with cellulose and hypolipidemic drug, advocating that chitosan is a potent agent for increasing HDL and lowering LDL blood cholesterol level and is more potent hypocholesterolemic agent than cellulose and the reference drug in this study. According to Park *et al* (2010)[6], nanopowder of chitosan reduced LDL by 55.7% and increased HDL by 16.5% in male rats, corroborating the present observation. According to Hsein and Meng (2005)[7], feeding 4% chitosan prepared from shrimp shell chitin was able to lower LDL level in hamsters, but had no effect on HDL level; however hamsters fed with chitosan showed higher ratio of HDL to TC than those fed with 4% cellulose. According to *Xu et al* (2007)[8], chitosan was able to decrease LDL level by 5% significance but the level of HDL remains unchanged. In this experiment, cellulose showed insignificant activity than chitosan for increasing HDL and decreasing LDL. Kritchevsky (1987)[9] also reported that cellulose is not hypocholesterolemic. Another finding by Ullrich (1987)[10]

suggests that insoluble dietary fibre like cellulose show better laxative effects with little change in plasma lipid levels in most studies.

The evident property of chitosan of reducing LDL level in the blood is due to its cationic nature and having affinity towards anionic molecules of fats and bile acids that prevents them from getting absorbed in the intestine but rather getting excreted out of the body. According to Ylitalo *et al* (2002)[11], the cationic chitosan binds with anionic carboxyl group of fatty acids and bile acids, it also interferes with emulsification of neutral lipids like cholesterol and other sterol by binding them with hydrophobic interaction, thus fat and cholesterol absorption from gastrointestinal tract is reduced. Another evidence of how chitosan reduces LDL level was given by Xu *et al* (2007)[8].According to them liver contains LDL receptors that are responsible for clearance of LDL from circulation, and chitosan increases hepatic LDL receptor mRNA expression in male rats as compared to cellulose.

The HDL increasing property of chitosan has also been proved in this study which follows the reverse cholesterol transport system. According to Singh (2016)[12], HDL serves as a chemical shuttle that transports excess cholesterol from peripheral tissues to the liver. Cholesterol undergoes esterification by lecithin-cholesterol acyltransferase (LCAT) to produce cholesterol ester, which results in the production of mature spherical HDL. Thus chitosan first decreases LDL level in the circulation thereby generating mature HDL in the liver.

This study indicated that the cationic biopolymer chitosan extracted from exoskeleton of freshwater crab Sartoriana spinigera leads to increased HDL level and decreased LDL level in blood cholesterol and has proved to be better than plant source cellulose and a synthetic hypolipidemic drug Ezetimibe proving that chitosan has hypocholesterolemic effect and is beneficial against cardiovascular diseases. Presently, when chemical and synthetic drugs are being used to control hypercholesterolemia, the proven effect of chitosan can become a boon for a sustainable society.

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